# Overcoming Visual Clutter in Vision Language Action Models via Concept-Gated Visual Distillation

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***Abstract*— Vision-Language-Action (VLA) models demonstrate impressive zero-shot generalization but degrade sharply in cluttered scenes, a failure we term the "Precision-Reasoning Gap." We hypothesize that this failure reflects background-induced feature dilution, in which semantically confusable clutter corrupts the visual grounding required for precise manipulation. To mitigate it, we propose Concept-Gated Visual Distillation (CGVD), a training-free, model-agnostic inference-time framework that parses the instruction into safe and distractor sets, applies a two-layer target refinement—cross-validation and spatial disambiguation—that explicitly penalizes false positives, and removes the remaining distractor regions via image inpainting, preserving background context and scene layout outside the edited regions. Compared against the unaugmented $\pi_0$ and GR00T policies in SimplerEnv, CGVD substantially improves robustness to semantic clutter—on *put spoon on towel* with 18 semantic distractors, $\pi_0$'s success rate rises from 43.0% to 77.5% (Table IV)—and, under descriptive compositional prompts, to attribute clutter. The protection is task- and distractor-type-conditional: on *put carrot on plate*, where contextual clutter aids the baseline policies, CGVD provides no benefit at any tested density and often significantly underperforms them. Runtime compositing adds a measured $\approx$16 ms (+5%) per control step after a one-off per-episode initialization of 0.7–3.3 s depending on scene complexity. These results support inference-time visual distillation as a practical defense against semantically confusable clutter for frozen VLA policies in static scenes.**

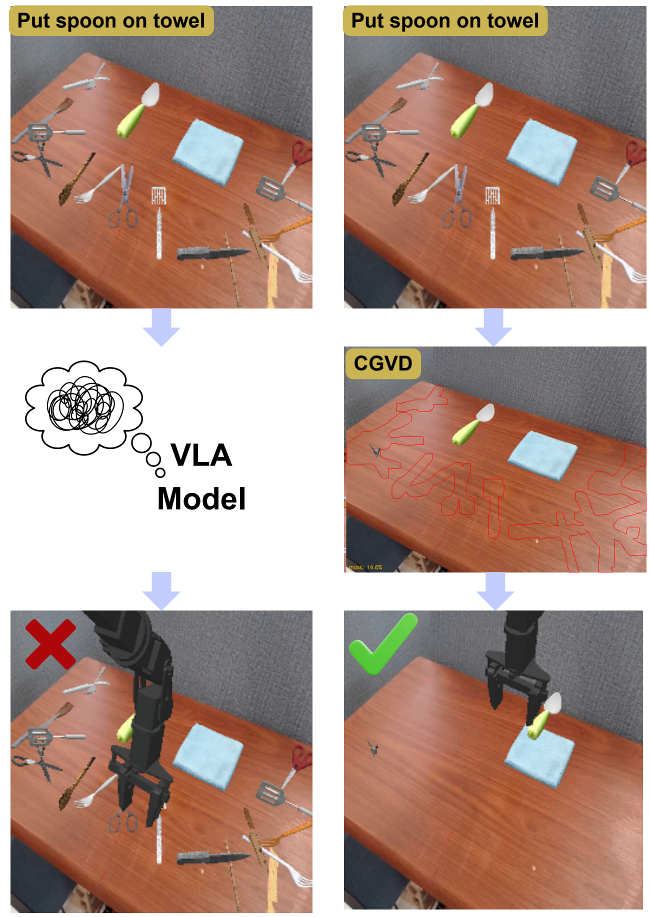


Fig. 1. Comparison of manipulation task execution in cluttered environments. While a standard VLA model (left) struggles with object confusion in a highly cluttered scene, our CGVD approach (right) successfully identifies and places the target object ("spoon") on the towel.

## I. INTRODUCTION

The pursuit of general-purpose robotic manipulation has been fundamentally accelerated by the advent of Vision-Language-Action (VLA) models [1]–[5]. By grounding large language models into robotic control policies, these architectures have demonstrated remarkable zero-shot generalization capabilities, enabling robots to follow open-vocabulary instructions like "Put spoon on towel" without task-specific training. These models promise a future where robots can operate in unstructured, human-centric environments.

However, a significant gap remains between the semantic reasoning capabilities of these models and their geometric precision in deployment conditions. While VLAs excel in curated, clutter-free environments, their performance degrades precipitously in the presence of visual clutter [6], [7] (Fig. 1). We term this phenomenon the Precision-Reasoning Gap: the model successfully identifies the target object conceptually, yet fails to manipulate it precisely. We hypothesize that attention corruption from surrounding distractors dilutes the latent representation used for spatial planning [8]; under this hypothesis, feature dilution would manifest as high-variance trajectories, hesitation near distractors, and ultimately manipulation failure. Our experiments (§IV) measure the end-to-end effect of removing distractors; they do not directly instrument the attention mechanism, so we present this account as a working hypothesis rather than an established mechanism.

Critically, this degradation is not uniform across distractor types. We observe that failure concentrates around distractors sharing visual or semantic properties with the target [6]. While VLAs may exhibit resilience to arbitrary clutter via large-scale pre-training, they remain brittle to semantically

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confusable objects; for example, a fork scattered near a target spoon triggers conflicting visual tokens within the same affordance category, causing the policy to attend to or even grasp the wrong object.

Existing approaches to mitigate clutter-induced failure fall into three primary categories. Adaptation methods, such as OBEYED-VLA [8], fine-tune attention adapters to focus on targets. While effective, this requires expensive, architecture-specific retraining and limits generalization to the fine-tuning distribution. Inference-time intervention methods, such as BYOVLA [9], use a VLM to identify distractors and a sensitivity probe to determine which to remove. This approach relies on external API calls (GPT-4o) and requires multiple VLA forward passes per region for its probe; its target protection is threshold-based, so the target can be modified if both the VLM and the sensitivity threshold fail to flag it. Training-time augmentation methods [10]–[12] generate diverse cluttered training data via generative models, improving robustness at the cost of retraining and without guarantees at deployment.

To bridge this gap without the cost of retraining, we propose Concept-Gated Visual Distillation (CGVD): a model-agnostic inference framework that leverages modern vision foundation models [13] to selectively restructure visual observations before they reach the VLA policy. CGVD parses the task instruction to identify target and anchor objects, segments both distractors and task-relevant entities independently, and uses set-theoretic subtraction to produce a distractor mask from which the target is excluded *conditional on correct safe-set segmentation*: the subtraction guarantees that no pixel segmented into the safe set is inpainted, but an object the segmenter fails to place in the safe set is not protected (§IV-D). Distractors are then replaced via inpainting [14], preserving the surrounding scene context.

Our contributions are as follows:

- **Concept-Gated Visual Distillation (CGVD):** We introduce a training-free, model-agnostic inference framework that selectively removes distractors from VLA observations via language-grounded segmentation and inpainting, while preserving scene context.
- **Interaction-Aware Masking Logic:** To overcome the inherent semantic confusion of open-set vision foundation models, which evaluate text prompts independently, we propose a set-theoretic cross-validation pipeline. This logic explicitly penalizes false positives and uses spatial disambiguation to isolate true targets from visually confusing distractors.
- **Systematic clutter-density evaluation:** We evaluate CGVD on two VLAs ($\pi_0$, GR00T) within the SimplerEnv benchmark over more than 20,000 episodes, comparing each policy with and without CGVD under matched seeds. CGVD prevents policy collapse under dense *semantic* clutter—reaching 77.5% success against the un-augmented policy's 43.0% ($\pi_0$, *put spoon on towel*, 18 semantic distractors; Table IV)—and improves adherence to compositional attribute prompts, while *underperforming* the baseline on a task where contextual clutter aids the policy (§IV-B). All comparisons in this paper are against the un-augmented policies; we do not empirically compare against competing intervention methods (§IV-A).

## II. Related Work

### A. Vision-Language-Action Models

The convergence of large language models with robotic control has yielded a new class of generalist policies known as Vision-Language-Action (VLA) models. Architectures such as RT-2 [2] and OpenVLA [1] leverage internet-scale vision-language pre-training to achieve remarkable zero-shot generalization to unseen objects and instructions. Similarly, policies like Octo [3] utilize transformer-based diffusion heads to aggregate behavior across diverse robot embodiments. While these models excel at general tasks with simple layouts, they often struggle with high-precision geometric grounding in cluttered scenes [6], [7]. Contributing factors include the static patch resolution inherent in Vision Transformer (ViT) backbones [15] and the absence of coarse-to-fine zooming mechanisms [16]; we hypothesize that these properties make such architectures susceptible to feature dilution [6], where background clutter competes with task-relevant signals for representational capacity.

### B. Vision Foundation Models for Robotic Perception

Recent advances in Vision Foundation Models (VFMs) have decoupled perception from specific task training. Architectures such as SAM 3 [13] and GroundingDINO [17] enable zero-shot object localization based on free-form text queries. In robotics, these tools have been primarily employed to *add* information: generating 3D value maps [18], constructing semantic maps [19], labeling datasets for offline training [20], or highlighting target regions via visual prompting [21]. Our work inverts this paradigm: we leverage the open-set discriminative power of VFMs to identify and *suppress* task-irrelevant regions. By explicitly subtracting task-irrelevant pixels, our framework acts—in an informal, analogical sense—as a semantic information bottleneck [22], removing clutter while preserving the geometric signals essential for the downstream VLA. Unlike [22], we impose no learned rate–distortion trade-off; the "bottleneck" here is a hard, hand-specified pixel-space mask.

### C. Robustness via Adaptation and Training-Time Augmentation

Addressing visual clutter has traditionally relied on domain randomization [23] or massive-scale pre-training on diverse datasets [24], [25], yet modern VLAs remain fragile to distributional shifts in cluttered scenes [26], [27]. Adaptation methods such as OBEYED-VLA [8] train specialized attention layers to recover geometric grounding, at the cost of architecture-specific retraining. Training-time augmentation methods generate cluttered or counterfactual training data with generative models: ROSIE [10] and GenAug [11] diversify backgrounds and objects, while NICE [12] performs pixel-space *scene surgery* on training data—the closest prior

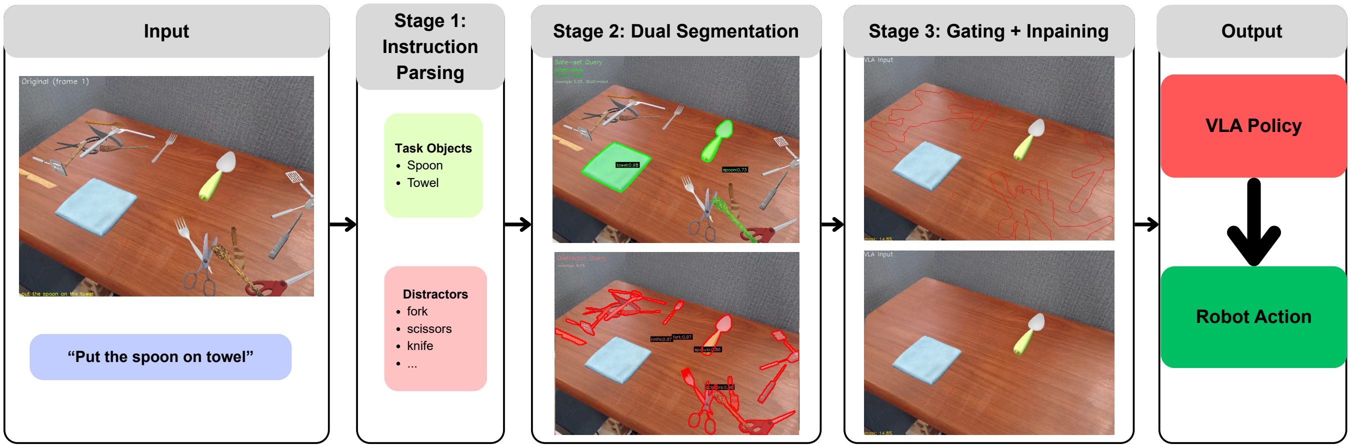


Fig. 2. Overview of the CGVD pipeline. **Stage 1:** The language instruction is parsed to extract a safe set (target, anchor) and a distractor set. **Stage 2:** SAM 3 segments both sets independently, producing a safe-set mask and a distractor mask via dual-channel segmentation. **Stage 3:** Set-theoretic gating subtracts the safe-set mask from the distractor mask, and LaMa inpaints the resulting regions to produce a distilled observation passed to the VLA policy.

to our core operation. CGVD performs a related pixel-space edit, but at *inference* time on a frozen policy: it requires no retraining and adapts to the current instruction, at the price of inheriting run-time perception risk that training-time methods amortize into the learned weights.

### D. Inference-Time Attention Intervention

A growing body of work attempts to correct VLA behavior during inference. Distracting Token Pruning (DTP) [28] detects and suppresses visual tokens that do not align with the text prompt, using a 'soft' pruning strategy in feature space. We conjecture that feature-space pruning struggles when distractors share semantic features with the target, whose token representations may already be entangled by early self-attention layers; our approach instead intervenes in pixel space, before tokenization. Unlike Visual Prompting [21], which adds information to guide attention, we utilize vision foundation models to remove information, preventing attention leakage to distractors. BYOVLA [9] also intervenes on the observation at run time; §IV-A states why no head-to-head comparison is reported here.

## III. Methodology

We present Concept-Gated Visual Distillation (CGVD), a training-free, model-agnostic inference-time framework that selectively removes distractor objects from a robot's visual observations while preserving task-relevant entities (Fig. 2). CGVD operates as a perception wrapper around any Vision-Language-Action (VLA) policy, requiring no fine-tuning or policy modification: given an instruction $l$ and observation $o_t \in \mathbb{R}^{H\times W\times 3}$, a distillation function $\phi$ produces a clean observation $\hat{o}_t = \phi(o_t, l)$ in which distractors are replaced with background, and the policy acts on $a_t = \pi(\hat{o}_t, l)$. Because $\phi$ interacts with $\pi$ only through the observation interface, it is agnostic to the policy architecture. The key insight is that the language instruction already specifies which objects matter; CGVD leverages this signal to gate visual content.

### A. Concept-Gated Decomposition

CGVD begins by parsing the language instruction $l$ to extract a target concept $c_{\text{tgt}}$ and an anchor concept $c_{\text{anc}}$. For instance, "put spoon on towel" yields $c_{\text{tgt}} = \texttt{spoon}$ and $c_{\text{anc}} = \texttt{towel}$. These concepts define two sets:

- The **safe set** $\mathscr{S} = \{c_{\text{tgt}}, c_{\text{anc}}\}$: entities that must remain visible. The robot arm is also always preserved, but it is handled through a separate proprioceptive channel (§III-E–III-F) rather than through the segmentation-based safe-set mask.
- The **distractor set** $\mathscr{D} = \{d_1, \ldots, d_K\}$: semantic categories to be treated as clutter (e.g., spatula, fork, knife).

The safe set is derived deterministically from the instruction. The distractor set is *not*: it is an externally supplied vocabulary. In our experiments, $\mathscr{D}$ is re-derived at every episode reset from the category labels of the objects the evaluation protocol actually spawned in that episode (§IV-A)—i.e., CGVD receives the exact ground-truth clutter inventory, per episode. At deployment, $\mathscr{D}$ would have to be supplied by the integrator as a site-specific vocabulary or proposed by an external model. Consequently, where prior work relies on a large language model to determine object relevance [9], our parsing is deterministic and requires no additional API *in this closed-vocabulary setting*; the two designs occupy different points on a coverage-versus-cost trade-off rather than one dominating the other.

### B. Text-Prompted Instance Segmentation

Both the safe set and distractor set are segmented from the observation using SAM 3 [13]. The two sets produce independent mask channels:

$$M_{\text{dist}} = \textstyle\bigcup_{d_k \in \mathscr{D}} \textsc{Seg}(o_t, d_k), \tag{1}$$

$$M_{\text{safe}} = \textsc{Seg}(o_t, c_{\text{tgt}}) \cup \textsc{Seg}(o_t, c_{\text{anc}}), \tag{2}$$

where $\textsc{Seg}(o_t, c)$ denotes the union of all instance masks returned by SAM 3 for concept $c$ in observation $o_t$. An important computational optimization is that the vision encoder is executed only once for the initialization frame ($t = 0$), with its masks reused across all frames of the episode.

### *C. Two-Layer Target Refinement*

A fundamental limitation of open-set segmentation models is that they evaluate text prompts independently. Consequently, a single-pass detection often suffers from semantic confusion, where visually similar distractors are misidentified as the target (e.g., a spatula yielding high confidence for the prompt "spoon"). To ensure the true target survives the distillation process, we introduce a two-layer refinement pipeline on the initial frame ($t = 0$).

**Layer 1: Cross-Validation.** To penalize distractors, we compute a genuineness score $g(s_i)$ for each target instance $s_i$. This measures the confidence differential between its safe-set and distractor identities:

$$g(s_i) = \sigma_{\text{safe}}(s_i) - \max_{\substack{d_j \in \mathscr{D} \\ \text{IoU}(s_i, d_j) > \eta}} \sigma_{\text{dist}}(d_j), \tag{3}$$

where $\sigma_{\text{safe}}$ and $\sigma_{\text{dist}}$ are object confidences, and $\eta$ is an IoU threshold (Table I). When no distractor satisfies $\text{IoU}(s_i, d_j) > \eta$—the common case for an isolated target—the maximum is defined as 0, so $g(s_i)$ reduces to $\sigma_{\text{safe}}(s_i)$. Genuine targets yield $g > 0$, while imposters yield $g < 0$. Negative values are explicitly preserved to actively penalize false positives in the next layer.

**Layer 2: Spatial Disambiguation.** Even after cross-validation, the target mask may contain fragmented artifacts or multiple disjoint physical objects. To isolate the correct entity, we evaluate each connected component $C_k$ using a composite score:

$$\text{score}(C_k) = \big(1 + g^*(C_k)\big) \cdot \sigma^*(C_k). \tag{4}$$

Here, $g^*(C_k)$ is maximum genuineness, and $\sigma^*(C_k)$ is peak safe-set confidence. These factors jointly favor genuine and high-confidence components. Only the top-scoring component is retained; the pipeline therefore assumes a *single* target instance per scene (§V).

To illustrate, consider a scene with a target `spoon` and a `spatula` distractor. If the model misidentifies the spatula as a "spoon" ($\sigma_{\text{safe}} = 0.6$) but correctly detects it as a "spatula" ($\sigma_{\text{dist}} = 0.9$), its genuineness drops to $-0.3$. Consequently, its Layer 2 composite score is heavily penalized ($(1 - 0.3) \cdot 0.6 = 0.42$). This allows the true spoon (which maintains a high positive genuineness) to outscore the imposter and be successfully isolated.

### *D. Concept-Gated Mask Composition*

The refined masks are combined into a final inpainting mask via set-theoretic gating:

$$M_{\text{inp}} = \text{dilate}(M_{\text{dist}}, r_d) \setminus \text{dilate}(M_{\text{safe}}, r_s), \tag{5}$$

where $r_d$ is a distractor dilation radius, and $r_s \geq r_d$ is a safe-set dilation radius that creates a protective buffer; the implementation enforces $r_s \geq r_d$ by construction, and both are 11 px in our configuration (Table I). All masks are binarized with a threshold of 0.5 before dilation to eliminate soft-value artifacts.

### *E. Clean Scene Generation via Inpainting*

We generate a single clean scene—an image of the workspace with distractors removed—by applying LaMa [14], a Fourier convolution-based inpainting model, to the initial frame. The inpainting mask is constructed as:

$$M_{\text{lama}} = M_{\text{inp}} \cup \text{dilate}(M_{\text{robot}}, r_e), \tag{6}$$

where $M_{\text{robot}}$ is the robot mask (obtained as described in §III-F) and $r_e$ is a robot dilation radius (Table I). Including the robot region lets LaMa synthesize the background *behind* the arm, so that the cached clean scene contains no stale robot pixels once the arm moves. A consequence is that scene content occluded by the arm at $t$=0 is synthesized rather than observed for the remainder of the episode (§V). LaMa fills the masked regions with photorealistic background texture. The clean scene is computed once per episode and cached for all subsequent frames.

### *F. Temporally Consistent Compositing*

At each timestep $t > 0$, the distilled observation is produced by alpha-blending the live camera frame with the cached clean scene:

$$\hat{o}_t = \alpha \odot \hat{o}_{\text{clean}} + (1 - \alpha) \odot o_t, \tag{7}$$

where the compositing mask $\alpha \in [0, 1]^{H \times W}$ is computed *once* during the $t$=0 warm-up and held fixed for the episode: the binarized, safe-set-gated distractor mask is feathered with a Gaussian blur ($\sigma$ in Table I), then clamped to 1 inside core distractor regions and to 0 inside the (dilated) target region, so that soft transitions occur only at region boundaries. To ensure that inpainting artifacts do not obscure the robot arm and compromise visual proprioception, we enforce a pixel-level overwrite of the robot onto the final composited image at every step; this is the only per-frame component of the compositing stage. In simulation, the robot mask is taken from SimplerEnv's renderer segmentation buffer (per-link actor IDs), which is pixel-perfect and free of frame-to-frame jitter. This mask is *simulation-privileged*: it has no direct real-world counterpart, and the cost of substituting a predicted (e.g., SAM 3) mask is not measured in this paper (§V). On real hardware, the natural replacement is a forward-kinematics projection of the robot model through the calibrated camera, which is deterministic, jitter-free by construction, and adds no perception cost; we identify this as the intended deployment path.

Two properties of this design should be stated explicitly. First, the distilled frame $\hat{o}_t$ is the *only* visual input the policy receives in our evaluation stack; no raw-observation channel reaches the policy. Second, the characteristic failure mode of observation editing is *false-positive erasure*: a physically present object that is misclassified as a distractor—or that enters the workspace after $t$=0, where the fixed $\alpha$ can blend it toward the cached background—is removed from the policy's view while remaining in the world. An arm commanded from such an observation can collide with an object it cannot see. Our evaluation is therefore scoped to

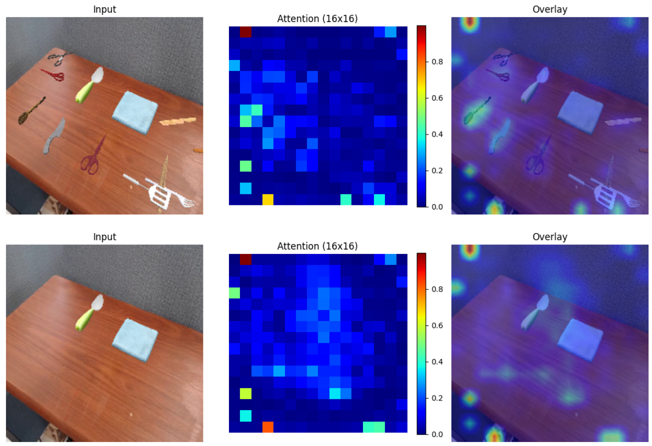


Fig. 3. **Qualitative illustration of the attention-repair hypothesis.** (Top) Under clutter, the baseline policy's attention is dispersed across distractors rather than the spoon. (Bottom) With CGVD, attention concentrates on the true target. Maps are produced by the evaluation harness's attention-capture instrument: hooks on every self-attention layer of $\pi_0$'s PaliGemma backbone recompute $\mathrm{softmax}(QK^\top/\sqrt{d})$; attention from the final query token onto the 256 SigLIP visual tokens (16×16 grid) is averaged over all layers and heads, captured at the first inference step of an episode (baseline: raw frame; CGVD: distilled frame). Because deleting a distractor's pixels necessarily deletes attention on those pixels, this figure illustrates the hypothesis but cannot by itself discriminate it from alternatives (§IV-B).

static workspaces with no human entry, a fixed camera, and a known clutter vocabulary; we do not claim validated operation outside this envelope.

## IV. Experiments

We evaluate CGVD across two VLA architectures on tabletop manipulation tasks with controlled distractor injection. Our experiments address three questions: (1) Does CGVD improve clutter robustness across different VLA architectures and tasks? (2) How does performance scale with distractor density? (3) How do individual CGVD components contribute at the operating point tested?

### A. Experimental Setup

**Environment.** We evaluate in SimplerEnv [29], a simulation benchmark with demonstrated real-to-sim correlation for VLA evaluation. SimplerEnv's correlation evidence supports *policy-level* transfer of our findings. It does not bound the perception-level risk of CGVD itself: SAM 3 [13] and LaMa [14] were trained on real imagery and are applied here to rendered frames—itself a domain shift—and two pipeline inputs are simulation-privileged (the renderer robot mask, §III-F, and the closed distractor vocabulary, §III-A). We therefore scope all conclusions to simulation and discuss transfer risk in §V.

All experiments use the WidowX robotic arm with a single fixed third-person camera, matching the Bridge dataset training setup. We test two Bridge dataset tasks: (1) *put spoon on towel* and (2) *put carrot on plate*.

**Policies.** $\pi_0$ refers to the open-source `open-pi-zero` reproduction of $\pi_0$ [5], Bridge checkpoint `bridge_beta`[1]; GR00T refers to `nvidia/GR00T-N1.6-bridge` [4]. Both are frozen; CGVD touches only their image input.

**Distractors.** Following the distractor-type framing of [6], we evaluate three types: (1) *Semantic*: objects with high semantic proximity and shared visual characteristics with the target; (2) *Random*: arbitrary objects with no semantic or visual similarity; and (3) *Attribute*: objects sharing the same category and form as the target but differing in physical properties. Distractor assets are drawn from RoboCasa [27] and the YCB [30] dataset and placed with collision-aware grid-based placement on a fixed 6 cm grid (one object per cell); when an object cannot be placed—cells exhausted at high densities—it is placed at a collision-free random position, or hidden off-scene if none exists, rather than the episode being dropped. The CGVD distractor vocabulary $\mathscr{D}$ is re-derived per episode from the category labels of the actually-spawned assets (e.g., `rc_fork_0` → "fork"), i.e., $\mathscr{D}$ coincides exactly with each episode's injected inventory (§III-A).

**Protocol.** We report success rate (SR) using SimplerEnv's built-in per-task success predicate, over episodes of 60 control steps. Each condition consists of 20 episodes per seed across 10 random seeds (200 episodes per task/distractor count; the attribute study of Table III uses 5 seeds, i.e. 100 episodes per cell), with matched seeds between the baseline and CGVD arms. The run seed $s \in \{0,\dots,9\}$ seeds the Python/NumPy/PyTorch RNGs (policy stochasticity); within a run, episode $e$ uses SimplerEnv's canonical initial layout $(s+e)$ mod 24 and a distractor-selection/placement seed $10^4 s+e$, so both arms see identical initial states and clutter layouts. Distractor counts are sampled at $\{0,4,8,10,14,18\}$. No episodes were excluded from analysis. We compare CGVD only against the un-augmented policies; no competing intervention method is run. BYOVLA [9] has a public implementation, while at the time of writing we found no public implementations of DTP [28] or OBEYED-VLA [8].

**Analysis.** For every condition we report the episode-level mean success rate and, as dispersion, the standard deviation of the 10 per-seed rates; differences carry 95% CIs computed from the per-seed paired deltas ($t_9$). Episode-level clustering by seed is negligible in these data (one-way ICC across all condition arms: median $-0.02$, maximum 0.06), so unpaired binomial intervals give similar widths; we report the seed-paired form because it matches the design. We treat the paper's headline contrast—$\pi_0$, *put spoon on towel*, 18 semantic distractors—as the single confirmatory comparison ($\Delta = +34.5$ pp, 95% CI $[+20.8, +48.2]$, paired $t$-test $p = 3\times10^{-4}$; Wilcoxon signed-rank $p = 0.002$); all other reported contrasts are exploratory and unadjusted, and deltas whose CI contains zero should be read as within run-to-run noise. The 0-distractor cells are shared between the semantic and random conditions (they are physically identical). The one condition that was executed twice end-to-end ($\pi_0$, *carrot*, random, 18 distractors) differed between batches by 3.0 pp (baseline) and 0.5 pp (CGVD), giving a direct run-to-run replication estimate.

**Implementation.** Table I lists all CGVD parameters.

[1]`github.com/allenzren/open-pi-zero`

TABLE I

**CGVD IMPLEMENTATION PARAMETERS** (HELD FIXED ACROSS ALL REPORTED EXPERIMENTS).

| Parameter | Value |
|---|---|
| SAM 3 checkpoint | `facebook/sam3` |
| LaMa checkpoint | big-lama (`simple-lama-inpainting`) |
| Safe-set (target/anchor) confidence threshold | 0.30 |
| Distractor confidence threshold | 0.20 |
| Cross-validation IoU threshold $\eta$ (Eq. 3) | 0.30 |
| Distractor dilation $r_d$ (Eq. 5) | 11 px |
| Safe-set dilation $r_s$ (Eq. 5) | 11 px ($= \max(5, r_d)$) |
| Robot dilation $r_e$ (Eq. 6) | 20 px |
| Compositing Gaussian $\sigma$ (Eq. 7) | 3.0 |
| Mask binarization threshold | 0.5 |
| Min. connected-component size (Layer 2) | 50 px |
| Safe-set warm-up frames | 1 |
| Clean-scene cache refresh | never (once per episode) |
| SAM 3 target prompts | instruction noun phrases (e.g., "spoon") |
| SAM 3 distractor prompts | category labels of $\mathscr{D}$ |

Values were selected on small pilot rollouts (primarily of the spoon task) during development and frozen before any reported evaluation run was executed (verified from version-control history), and were then held constant across all tasks, policies, and distractor conditions. Code, configuration, and evaluation scripts are available at `github.com/awd1779/pi0`.

### B. Main Results

Figure 4 illustrates the task success rates as the number of distractors increases from 0 to 18; Table II reports the corresponding numbers with per-seed dispersion and paired CIs. We observe distinct trends across the evaluated VLA models ($\pi_0$ and GR00T) depending on the nature of the task and clutter.

On *put spoon on towel*, baseline performance degrades precipitously under semantically confusing clutter, and CGVD prevents this collapse: $\pi_0$ gains $+14.0$ to $+34.5$ pp for $n \geq 4$ and GR00T gains $+11.0$ to $+13.5$ pp for $n \geq 8$, with every paired CI excluding zero. The gap widens with density: the per-seed slope of the CGVD$-$baseline difference is $+1.54$ pp per distractor for $\pi_0$ ($t(9)$=5.54, $p$<0.001) and $+0.74$ for GR00T ($t(9)$=2.77, $p$=0.02). Random clutter on this task shows the same direction at high density ($\pi_0$: $+13.0$ $[+2.0, +24.0]$; GR00T: $+17.0$ $[+7.6, +26.4]$ at 18 distractors), so on this task the benefit is not exclusive to semantic clutter.

Conversely, on *put carrot on plate*, the baseline policies exhibit a performance increase at moderate distractor densities before degrading. This may arise because a moderate amount of contextual clutter better aligns with the object-rich scenes present in their large-scale pre-training distributions, providing visual anchors for reasoning.

On this task CGVD never significantly outperforms the baseline at any density and significantly underperforms it at several: $\pi_0$'s semantic deltas span $-4.0$ to $-9.0$ pp (CIs exclude zero at $n$=10 and $n$=18), and GR00T loses $-9.5$ to $-14.5$ pp at moderate densities under both clutter types. This suggests that when a task naturally benefits from contextual clutter, aggressively masking the scene deprives the VLA of useful environmental reasoning; aggressive inpainting can also introduce generative artifacts that disrupt the spatial geometry. Therefore, while CGVD provides critical protection against adversarial semantic distractors, it trades off performance in scenarios where background objects serve as beneficial visual anchors: the intervention is *task- and distractor-type-conditional*, not universally beneficial.

Figure 3 qualitatively illustrates our working hypothesis for the semantic-clutter effect: with CGVD, the policy's attention concentrates on the target. We emphasize that this observation is entailed by the intervention itself (removing a distractor's pixels removes attention on them) and that no attention statistic is measured in this paper; alternative explanations—such as CGVD restoring the observation toward the clutter-sparse Bridge training distribution, or simply deleting the wrong-grasp option—are consistent with our data and are not ruled out. The mean-color-fill ablation (§IV-D), in which removing the *same* semantic content with a cruder fill forfeits most of the gain, indicates that fill appearance, not distractor semantics alone, carries a large share of the effect.

### C. Fine-Grained Semantic Grounding

The precision-reasoning gap is most acute when distractors share a semantic class with the target but differ in specific attributes. In standard VLAs, visual tokens for complex queries like "Put spoon with green handle on towel" are often reduced to a bag-of-words, causing the policy to entangle the target with fully green objects or ignore modifiers entirely. To evaluate this, we use the *Attribute* distractor type (§IV-A): the target is a green-handled spoon and distractors are same-category spoons differing in color or material. We evaluate the baseline and CGVD across 0–4 attribute distractors using two prompt structures: Simple ("Put green spoon on towel") and Complex ("Put spoon with green handle on towel"). CGVD receives the same instruction as the policy; its SAM 3 target prompt is the instruction's attribute phrase ("green spoon" / "spoon with green handle"). Note the coverage asymmetry: the attribute type is evaluated on one task with $\pi_0$ only, at 0–4 distractors, whereas Fig. 4 covers the semantic and random types on both tasks and both policies at 0–18 distractors. Generality claims for attribute clutter are correspondingly scoped.

Table III reveals a two-sided dependence on descriptive density. On the baseline side, performance with simple adjective-noun prompts degrades only mildly with clutter (87.0% $\rightarrow$ 75.0%), but collapses on compositional queries (86.0% $\rightarrow$ 57.0%), confirming the bag-of-words brittleness. On the CGVD side, the complex prompt is what unlocks the benefit: CGVD holds a stable floor under the compositional query (73.0% at 4 distractors, $\Delta = +16.0$, the one delta in this table whose CI excludes zero), because SAM 3's open-set grounding exploits the rich attribute phrase to isolate the correct instance. With the *simple* prompt, by contrast, CGVD provides no reliable benefit (deltas $-5.0$ to $+3.0$, all within noise): the terse phrase "green spoon" does not give the segmenter enough descriptive signal to separate the target from attribute-conflicting spoons. CGVD thus converts

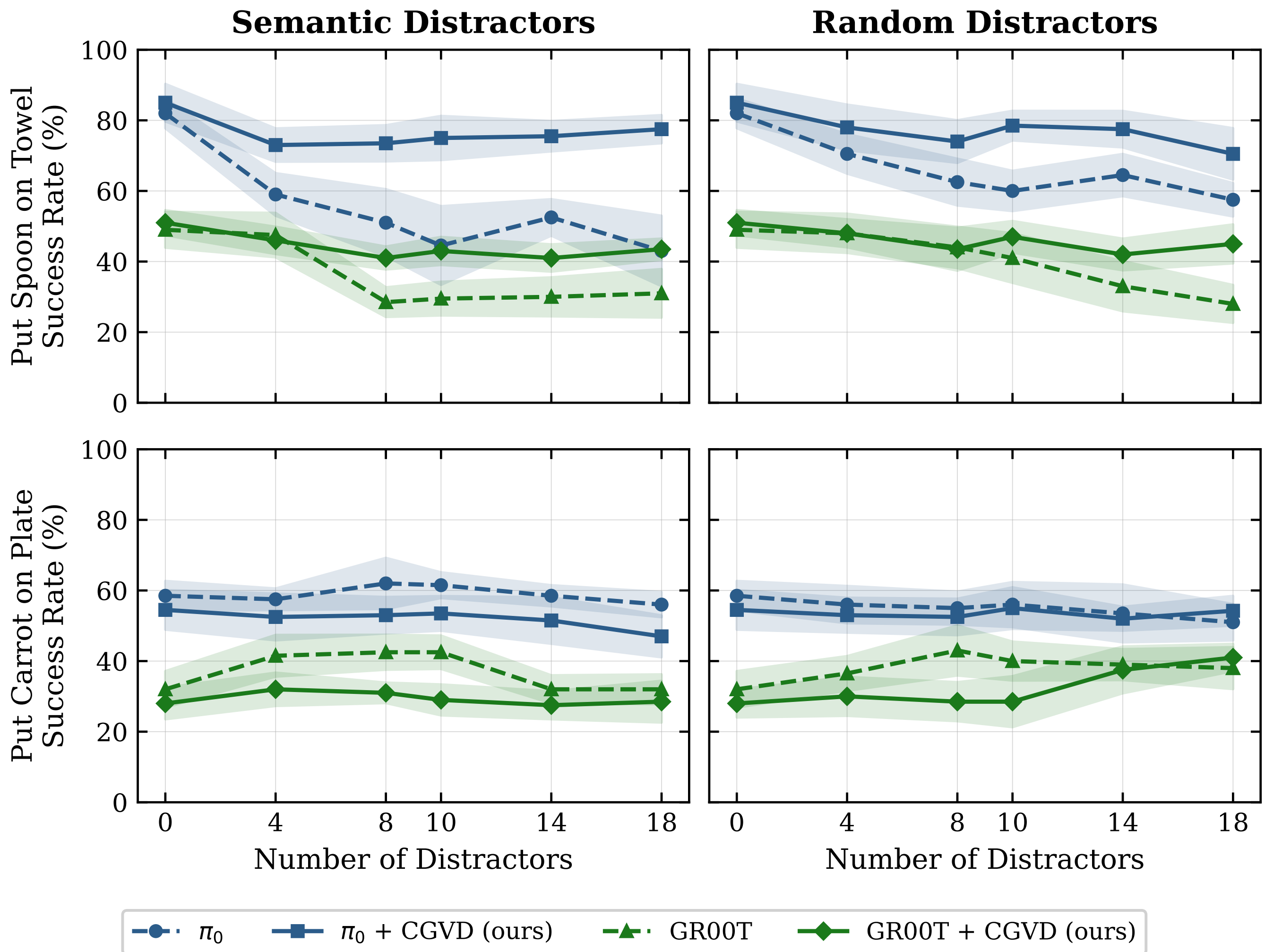


Fig. 4. Success rate vs. number of distractors. **Left:** semantic distractors. **Right:** random distractors. **Top:** *put spoon on towel*. **Bottom:** *put carrot on plate*. Dashed lines represent the baseline VLA; solid lines represent +CGVD. Colors denote specific model architectures. Each data point is the mean success rate over 200 evaluation rollouts (20 episodes $\times$ 10 matched seeds) at distractor counts $\{0, 4, 8, 10, 14, 18\}$, totaling 19,200 episodes for the results visualized in this figure. The 0-distractor points are shared between the semantic and random panels. Numeric values with per-seed dispersion and paired CIs are given in Table II.

TABLE II

MAIN RESULTS (NUMERIC). SUCCESS RATE (%), MEAN $\pm$ SD OVER THE 10 MATCHED SEEDS (20 EPISODES PER SEED; 200 EPISODES PER CELL). $\Delta$ IS THE CGVD $-$ BASELINE DIFFERENCE WITH A 95% CI FROM THE PER-SEED PAIRED DELTAS ($t_9$). BOLD MARKS DELTAS WHOSE PAIRED CI EXCLUDES ZERO. THESE ARE THE DATA PLOTTED IN FIG. 4.

| | | **Semantic distractors** | | | | | | **Random distractors** | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | $\pi_0$ | | | GR00T | | | $\pi_0$ | | | GR00T | | |
| **Task** | **#D** | base | +CGVD | Δ [95% CI] | base | +CGVD | Δ [95% CI] | base | +CGVD | Δ [95% CI] | base | +CGVD | Δ [95% CI] |
| *spoon on towel* | 0 | 82.0 | 85.0 | +3.0 [-5.6,+11.6] | 49.0 | 51.0 | +2.0 [-6.1,+10.1] | 82.0 | 85.0 | +3.0 [-5.6,+11.6] | 49.0 | 51.0 | +2.0 [-6.1,+10.1] |
| | 4 | 59.0 | 73.0 | **+14.0 [+5.4,+22.6]** | 47.5 | 46.0 | -1.5 [-8.7,+5.7] | 70.5 | 78.0 | +7.5 [-0.8,+15.8] | 48.0 | 48.0 | +0.0 [-7.0,+7.0] |
| | 8 | 51.0 | 73.5 | **+22.5 [+10.7,+34.3]** | 28.5 | 41.0 | **+12.5 [+6.4,+18.6]** | 62.5 | 74.0 | **+11.5 [+1.7,+21.3]** | 44.0 | 43.5 | -0.5 [-12.5,+11.5] |
| | 10 | 49.0 | 72.5 | **+23.5 [+12.8,+34.2]** | 29.5 | 43.0 | **+13.5 [+6.1,+20.9]** | 60.0 | 78.5 | **+18.5 [+10.2,+26.8]** | 41.0 | 47.0 | +6.0 [-3.0,+15.0] |
| | 14 | 52.5 | 75.5 | **+23.0 [+14.7,+31.3]** | 30.0 | 41.0 | **+11.0 [+3.3,+18.7]** | 64.5 | 77.5 | **+13.0 [+1.7,+24.3]** | 33.0 | 42.0 | +9.0 [-0.9,+18.9] |
| | 18 | 43.0 | 77.5 | **+34.5 [+20.8,+48.2]** | 31.0 | 43.5 | **+12.5 [+4.0,+21.0]** | 57.5 | 70.5 | **+13.0 [+2.0,+24.0]** | 28.0 | 45.0 | **+17.0 [+7.6,+26.4]** |
| *carrot on plate* | 0 | 58.5 | 54.5 | -4.0 [-11.7,+3.7] | 32.0 | 28.0 | -4.0 [-12.2,+4.2] | 58.5 | 54.5 | -4.0 [-11.7,+3.7] | 32.0 | 28.0 | -4.0 [-12.1,+4.1] |
| | 4 | 57.5 | 52.5 | -5.0 [-11.5,+1.5] | 41.5 | 32.0 | **-9.5 [-12.1,-6.9]** | 56.0 | 53.0 | -3.0 [-10.8,+4.8] | 36.5 | 30.0 | -6.5 [-16.5,+3.5] |
| | 8 | 62.0 | 53.0 | -9.0 [-21.1,+3.1] | 42.5 | 31.0 | **-11.5 [-17.1,-5.9]** | 55.0 | 52.5 | -2.5 [-10.8,+5.8] | 43.0 | 28.5 | **-14.5 [-21.1,-7.9]** |
| | 10 | 61.5 | 53.5 | **-8.0 [-15.0,-1.0]** | 42.5 | 29.0 | **-13.5 [-20.7,-6.3]** | 56.0 | 55.0 | -1.0 [-10.7,+8.7] | 40.0 | 28.5 | **-11.5 [-18.7,-4.3]** |
| | 14 | 58.5 | 51.5 | -7.0 [-17.4,+3.4] | 32.0 | 27.5 | -4.5 [-9.7,+0.7] | 53.5 | 52.0 | -1.5 [-13.8,+10.8] | 39.0 | 37.5 | -1.5 [-9.2,+6.2] |
| | 18 | 56.0 | 47.0 | **-9.0 [-17.7,-0.3]** | 32.0 | 28.5 | -3.5 [-10.0,+3.0] | 49.5 | 54.0 | +4.5 [-6.9,+15.9] | 38.0 | 41.0 | +3.0 [-3.6,+9.6] |

descriptive density into robustness; it does not rescue a prompt that lacks it. Given the 5-seed sample, we read the per-cell values in this table as descriptive.

### *D. Ablation Studies*

To examine CGVD's structural components, we ablate the pipeline at a single operating point: *put spoon on towel*,

TABLE III

**ATTRIBUTE DISTRACTOR SENSITIVITY.** SUCCESS RATES ON *put spoon on towel* WITH ATTRIBUTE DISTRACTORS ($\pi_0$ ONLY), 100 EPISODES PER CELL (5 MATCHED SEEDS × 20 EPISODES). Δ CARRIES A 95% CI FROM PER-SEED PAIRED DELTAS; BOLD MARKS DELTAS WHOSE CI EXCLUDES ZERO. DELTAS WHOSE CI CONTAINS ZERO ARE WITHIN RUN-TO-RUN NOISE.

| | Simple Prompt | | | Complex Prompt | | |
|---|---|---|---|---|---|---|
| # Distractors | $\pi_0$ | +CGVD | Δ [95% CI] | $\pi_0$ | +CGVD | Δ [95% CI] |
| 0 | 87.0 | 85.0 | -2.0 [-13.3,+9.3] | 86.0 | 87.0 | +1.0 [-5.8,+7.8] |
| 1 | 79.0 | 80.0 | +1.0 [-7.1,+9.1] | 74.0 | 69.0 | -5.0 [-26.9,+16.9] |
| 2 | 76.0 | 79.0 | +3.0 [-14.9,+20.9] | 69.0 | 77.0 | +8.0 [-9.9,+25.9] |
| 3 | 77.0 | 75.0 | -2.0 [-20.4,+16.4] | 64.0 | 74.0 | +10.0 [-7.6,+27.6] |
| 4 | 75.0 | 70.0 | -5.0 [-24.6,+14.6] | 57.0 | 73.0 | **+16.0 [+1.2,+30.8]** |

TABLE IV

**ABLATION STUDY** ON *put spoon on towel* ($\pi_0$, 18 SEMANTIC DISTRACTORS). EACH ROW REMOVES OR SUBSTITUTES ONE COMPONENT OF THE FULL CGVD PIPELINE; 200 EPISODES (20 EPISODES × 10 MATCHED SEEDS) PER ROW. SR IS MEAN ± SD OVER PER-SEED RATES; Δ VS. THE FULL PIPELINE CARRIES A 95% CI FROM PER-SEED PAIRED DELTAS. THE BASELINE AND FULL-PIPELINE ROWS REUSE THE CORRESPONDING FIG. 4 RUNS AT THIS CONDITION. THE ROBOT-MASK EFFECT IS NOT DISTINGUISHABLE FROM ZERO AT THIS SAMPLE SIZE.

| Configuration | SR (%) | Δ [95% CI] |
|---|---|---|
| Baseline (no CGVD) | $43.0 \pm 16.0$ | $-34.5$ $[-48.2, -20.8]$ |
| CGVD (full pipeline) | $\mathbf{77.5} \pm 6.3$ | — |
| LaMa → mean-color fill | $56.5 \pm 16.0$ | $-21.0$ $[-35.3, -6.7]$ |
| – Two-layer target refinement | $65.0 \pm 8.8$ | $-12.5$ $[-20.3, -4.7]$ |
| – Robot mask protection | $73.0 \pm 7.1$ | $-4.5$ $[-10.4, +1.4]$ |

$\pi_0$, 18 semantic distractors (Table IV). We use $\pi_0$ as the base policy because it achieves higher overall success rates, providing headroom in which component contributions can be resolved. The attributions below are therefore claims about this operating point only; our own main results (§IV-B) show that the pipeline's net effect reverses in other regimes, so component contributions should not be assumed to transfer.

Removing the Two-Layer Target Refinement reduces SR from 77.5% to 65.0%. Without cross-validation, the segmentation model cannot distinguish true targets from visually similar distractors, causing genuine targets to be erroneously inpainted out of the scene—i.e., CGVD's own target protection fails when the refinement is absent, which is why we describe the protection as conditional (§III-A). To quantify the residual failure rate of the *full* pipeline, we audited a fresh 200-episode re-execution of the headline condition against the renderer's ground-truth target mask: the final inpainting mask overlapped the target at $t$=0 in 4 of 200 episodes (2.0%), and in all four cases the target was erased essentially completely. The protection is therefore strong but probabilistic in practice, differing from sensitivity-probe approaches in failure *rate*, not failure *mode*. (This re-execution also provides a replication of the headline cell: 76.0% vs. the original 77.5%.) Substituting a mean-color fill for LaMa incurs the largest single drop (to 56.5%), even though the masks—and thus the removed semantic content—are identical. We hypothesize that the stark, unnatural region boundaries act like out-of-distribution patches to the VLA's ViT backbone, disrupting planning; we have not measured this mechanism. Notably, this row implies that a majority of CGVD's gain at this operating point depends on *how* the removed regions are filled rather than on the removal alone. Finally, removing Robot Mask Protection reduces SR to 73.0%; this $-4.5$ pp effect is not statistically distinguishable from zero at our sample size (95% CI $[-10.4, +1.4]$), so we treat robot-mask protection as qualitatively motivated—without it, the compositing mask occasionally and visibly occludes the robot arm—rather than as a measured gain.

### *E. Latency Analysis*

CGVD concentrates its expensive operations (SAM 3, LaMa) on the initialization frame ($t = 0$); for $t > 0$ the system performs image compositing against the cached clean scene. Table V reports timing measured on an NVIDIA A10G (23 GB) across the evaluation runs. The policy forward pass is unaffected ($318.1 \pm 6.6$ ms baseline vs. $318.5 \pm 5.5$ ms under CGVD), and per-frame compositing adds $15.6 \pm 0.4$ ms, i.e. ≈+5% per control step (≈3.14 → 3.00 Hz). Initialization is a one-off per-episode cost that grows with the number of segmentation prompts: $0.68 \pm 0.29$ s in the attribute scenes (≤5 prompts) and ≈3.3 s in the 18-distractor scenes (derived from episode wall time), delaying the first action of each episode. The overhead is modest for the quasi-static tabletop tasks evaluated here; the startup delay remains a real cost for time-critical settings.

TABLE V

**SYSTEM LATENCY**, MEASURED ON AN NVIDIA A10G ACROSS THE EVALUATION RUNS (MEAN ± SD OVER EPISODES; $n$ IN PARENTHESES). CGVD CONCENTRATES SEGMENTATION AND INPAINTING AT $t$=0; RUNTIME COMPOSITING ADDS ≈5% PER STEP.

| Quantity | Measured |
|---|---|
| Policy forward / step, baseline | $318.1 \pm 6.6$ ms (500 ep.) |
| Policy forward / step, with CGVD | $318.5 \pm 5.5$ ms (1,100 ep.) |
| CGVD compositing / frame ($t > 0$) | $15.6 \pm 0.4$ ms (500 ep.) |
| Initialization ($t = 0$), ≤5 prompts | $0.68 \pm 0.29$ s (500 ep.) |
| Initialization ($t = 0$), 18-distractor scenes | ≈3.3 s (wall-time derived) |

## V. LIMITATIONS

CGVD's validity envelope is defined by the following assumptions and scope limits.

**Static scene.** The clean scene is cached after the initialization frame and $\alpha$ is fixed (§III-F). If a distractor is moved, the cached background desynchronizes from the physical scene; more importantly, an object that *enters* the workspace after $t$=0 can be blended toward the cached background and hidden from the policy while physically present—a hazard, not merely an accuracy loss, in any workspace that is not access-controlled. Continuous re-segmentation would remove this assumption but is currently prohibitive at control rate, making the cached design a practical trade-off for static, human-excluded workspaces.

**Contextually informative clutter.** As §IV-B shows, on *put carrot on plate* CGVD never outperforms the baseline and significantly underperforms it at several densities (up to $-14.5$ pp for GR00T), where clutter appears to serve as a useful visual anchor; aggressive inpainting can also introduce generative artifacts. The intervention should be understood as task- and distractor-type-conditional, and deciding *when* to distill is open future work.

**Simulation-only evaluation and privileged inputs.** No real-robot experiment is reported. Two pipeline inputs are simulation-privileged: the renderer robot mask (§III-F), whose ablation row in Table IV measures the value of having protection but not the cost of *deriving* it without the simulator, and the closed distractor vocabulary $\mathscr{D}$ (§III-A), which coincides with the injected clutter inventory. Sim-to-real risk is therefore not limited to policy transfer: SAM 3 recall over $\mathscr{D}$, LaMa fill fidelity on real textured scenes (the largest single factor in Table IV), and per-frame robot masking are all unvalidated on hardware.

**Single target instance.** Layer 2 retains only the top-scoring component (§III-C); a scene legitimately containing two instances of the target concept would have one of them inpainted away.

**Occlusion fabrication and in-flight blending.** Because Eq. 6 inpaints the robot region at $t=0$, whatever the arm occluded in the initialization frame is synthesized, not observed, in the cached clean scene for the entire episode. Relatedly, the compositing mask $\alpha$ protects the target only at its $t=0$ location: auditing the headline condition against ground truth, the *moving* target's pixels overlap previously inpainted regions by more than 10% in 35% of episodes at some point during execution, exposing the grasped object to partial blending toward the cached background.

**Startup latency.** The single-frame initialization introduces a 0.7–3.3 s delay (scene-complexity dependent) before the first action of each episode (Table V).

## VI. Conclusion

In this paper, we introduced Concept-Gated Visual Distillation (CGVD), a training-free inference framework designed to mitigate the Precision-Reasoning Gap in VLA models. By leveraging language-grounded segmentation and inpainting, CGVD isolates target objects and suppresses semantic distractors without architectural modification. Across more than 20,000 SimplerEnv episodes with $\pi_0$ and GR00T, CGVD prevents performance collapse under dense semantic clutter and improves adherence to compositional attribute prompts, relative to the un-augmented policies. The benefit is conditional: on a task where contextual clutter aids the policy, CGVD never outperforms and often significantly underperforms the baseline, while its measured per-step overhead is small ($\approx$+5%). Bounded by static-background and closed-vocabulary assumptions, these results position inference-time visual distillation as a practical defense against semantically confusable clutter in static scenes. Future work will explore real-time mask updating for interactive clutter, intrusion detection over the compositing regions, open-vocabulary construction of the distractor set, and criteria for when distillation should be enabled at all.

## References


[1] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair *et al.*, "Openvla: An open-source vision-language-action model," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, 2024.

[2] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski *et al.*, "Rt-2: Vision-language-action models transfer web knowledge to robotic control," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, 2023.

[3] D. Ghosh, H. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari *et al.*, "Octo: An open-source generalist robot policy," in *Proceedings of Robotics: Science and Systems (RSS)*, 2024.

[4] J. Bjorck, F. Castaneda, N. Cherniadev, X. Da, R. Ding, L. Fan *et al.*, "Gr00t n1: An open foundation model for generalist humanoid robots," *arXiv preprint arXiv:2503.14734*, 2025.

[5] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn *et al.*, "$\pi_0$: A vision-language-action flow model for general robot control," in *Robotics: Science and Systems (RSS)*, 2025.

[6] A. Rasouli, M. Alban, S. Pakdamansavoji, Z. Li, Z. Zhang, A. Wu *et al.*, "Distracted robot: How visual clutter undermine robotic manipulation," *arXiv preprint arXiv:2511.22780*, 2025.

[7] H. Liu, J. Long, J. Wu, J. Hou, H. Tang, T. Jiang *et al.*, "Eva-VLA: Evaluating vision-language-action models' robustness under real-world physical variations," *arXiv preprint arXiv:2509.18953*, 2025.

[8] K. Vo, T. Hanyu, Y. Ikebe, T. T. Pham, N. Chung, M. N. Vu *et al.*, "Clutter-robust vision-language-action models through object-centric and geometry grounding," *arXiv preprint arXiv:2512.22519*, 2025.

[9] A. J. Hancock, A. Z. Ren, and A. Majumdar, "Run-time observation interventions make vision-language-action models more visually robust," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[10] T. Yu, T. Xiao, A. Stone, J. Tompson, A. Brohan, S. Wang *et al.*, "Scaling robot learning with semantically imagined experience," in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.

[11] Q. Chen, S. Kiami, A. Gupta, and V. Kumar, "GenAug: Retargeting behaviors to unseen situations via generative augmentation," in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.

[12] S. Pakdamansavoji, M. Pourkeshavarz, A. Sigal, Z. Li, R. H. Yang, and A. Rasouli, "Improving robotic manipulation robustness via NICE scene surgery," *arXiv preprint arXiv:2511.22777*, 2025.

[13] N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris *et al.*, "Sam 3: Segment anything with concepts," *arXiv preprint arXiv:2511.16719*, 2025.

[14] R. Suvorov, E. Logacheva, A. Mashikhin, A. Remizova, A. Ashukha, A. Silvestrov *et al.*, "Resolution-robust large mask inpainting with fourier convolutions," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2022, pp. 2149–2159.

[15] M. Raghu, T. Unterthiner, S. Kornblith, C. Zhang, and A. Dosovitskiy, "Do vision transformers see like convolutional neural networks?" in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[16] S. James, K. Wada, T. Laidlow, and A. J. Davison, "Coarse-to-fine Q-Attention: efficient learning for visual robotic manipulation via discretisation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[17] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang *et al.*, "Grounding dino: Marrying dino with grounded pre-training for open-set object detection," in *European Conference on Computer Vision (ECCV)*, 2024.

[18] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, "Voxposer: Composable 3d value maps for robotic manipulation with language models," in *Conference on Robot Learning (CoRL)*, 2023.

[19] K. M. Jatavallabhula, A. Kuwajerwala, Q. Gu, M. Omama, T. Chen, A. Maalouf *et al.*, "Conceptfusion: Open-set multimodal 3d mapping," in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.

[20] Y. Wang, Z. Xian, F. Chen, T.-H. Wang, Y. Wang, K. Fragkiadaki *et al.*, "Robogen: Towards unleashing infinite data for automated robot learning via generative simulation," in *International Conference on Machine Learning (ICML)*, 2024.

[21] Y. Li, Z. Gong, H. Li, X. Huang, H. Kang, G. Bai *et al.*, "Robotic visual instruction," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 12 155–12 165.

[22] S. Bai, W. Zhou, P. Ding, W. Zhao, D. Wang, and B. Chen, "Rethinking latent redundancy in behavior cloning: an information bottleneck approach for robot manipulation," in *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR, 2025, pp. 2560–2580.

[23] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017.

[24] H. Walke, K. Black, A. Lee, M. J. Kim, M. Du, C. Zheng *et al.*, "Bridgedata v2: A dataset for robot learning at scale," in *Conference on Robot Learning (CoRL)*, 2023.

[25] Open X-Embodiment Collaboration, "Open x-embodiment: Robotic learning datasets and rt-x models," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

[26] W. Pumacay, I. Singh, J. Duan, R. Krishna, J. Thomason, and D. Fox, "The colosseum: A benchmark for evaluating generalization for robotic manipulation," in *Robotics: Science and Systems (RSS)*, 2024.

[27] S. Nasiriany, A. Maddukuri, L. Zhang, A. Parikh, A. Lo, A. Joshi *et al.*, "Robocasa: Large-scale simulation of everyday tasks for generalist robots," in *Robotics: Science and Systems (RSS)*, 2024.

[28] C. Li, J. Liu, B. Li, B. Gao, Y. Yuan, Y. He *et al.*, "DTP: A simple yet effective distracting token pruning framework for vision-language action models," *arXiv preprint arXiv:2601.16065*, 2026.

[29] X. Li, K. Hsu, J. Gu, K. Pertsch, O. Mees, H. R. Walke *et al.*, "Evaluating real-world robot manipulation policies in simulation," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024.

[30] B. Calli, A. Singh, A. Walsman, S. S. Srinivasa, P. Abbeel, and A. M. Dollar, "The YCB object and model set: Towards common benchmarks for manipulation research," in *2015 International Conference on Advanced Robotics (ICAR)*. IEEE, 2015, pp. 510–517.